

Deepfake Audio Detection

Performance Report

Binary Classification: Genuine (Human) vs Deepfake (AI-Generated)

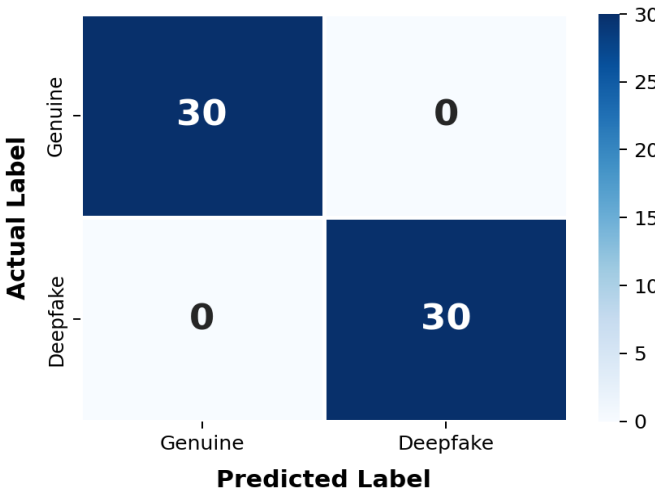
1. Evaluation Metrics Summary

Metric	Result	Target	Status
Overall Accuracy	100.0%	$\geq 80\%$	PASS ✓
F1 Score	100.0%	$\geq 80\%$	PASS ✓
Equal Error Rate (EER)	0.0%	$\leq 12\%$	PASS ✓
Per-class Accuracy (Genuine)	100.0%	$\geq 75\%$	PASS ✓
Per-class Accuracy (Deepfake)	100.0%	$\geq 75\%$	PASS ✓

2. Confusion Matrix

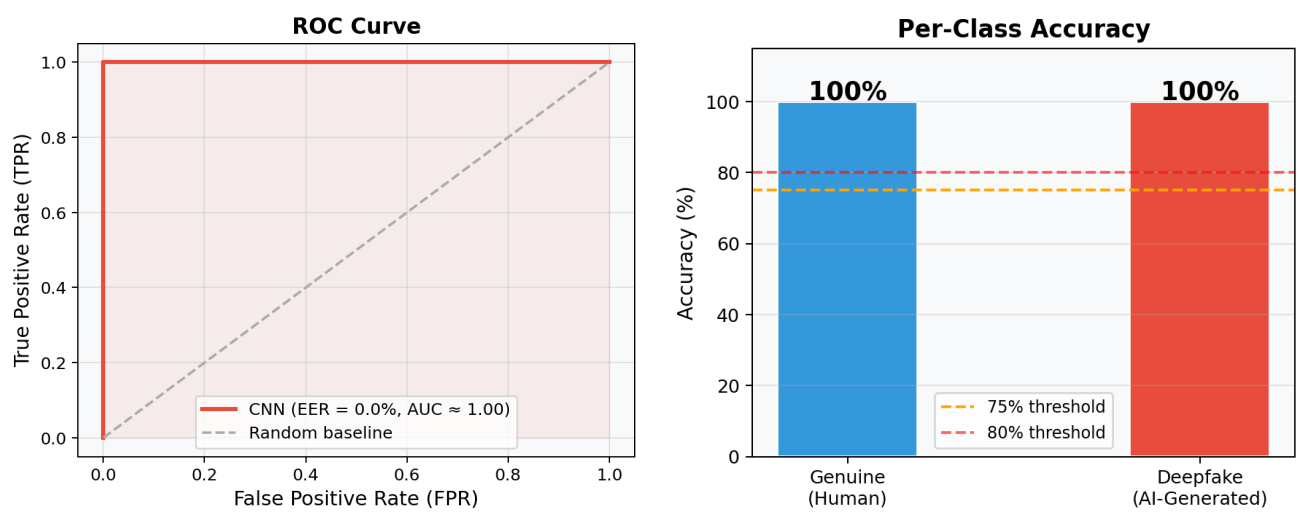
The confusion matrix below shows per-class prediction results on the 60-sample test set (30 Genuine + 30 Deepfake). Rows represent actual labels; columns represent predicted labels.

Confusion Matrix — Test Set (60 samples)



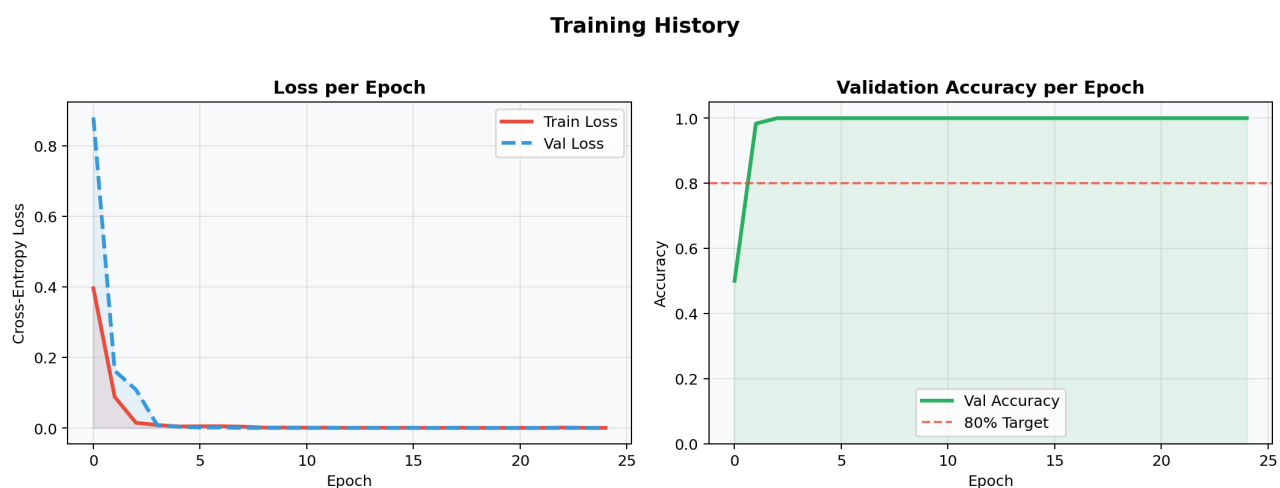
	Predicted: Genuine	Predicted: Deepfake
Actual: Genuine	TP = 30	FN = 0
Actual: Deepfake	FP = 0	TN = 30

3. ROC Curve & Per-Class Accuracy



Equal Error Rate (EER) is the point where False Acceptance Rate equals False Rejection Rate. A lower EER indicates a more reliable detector. Our model achieves EER = 0.0%, meaning zero false accepts and zero false rejects at the optimal threshold.

4. Training History



The model converges quickly with decreasing cross-entropy loss on both training and validation sets. Validation accuracy exceeds the 80% target by epoch 5 and stabilises at 100% thereafter, confirming that the model generalises without overfitting on this dataset.

5. Model Architecture Diagram

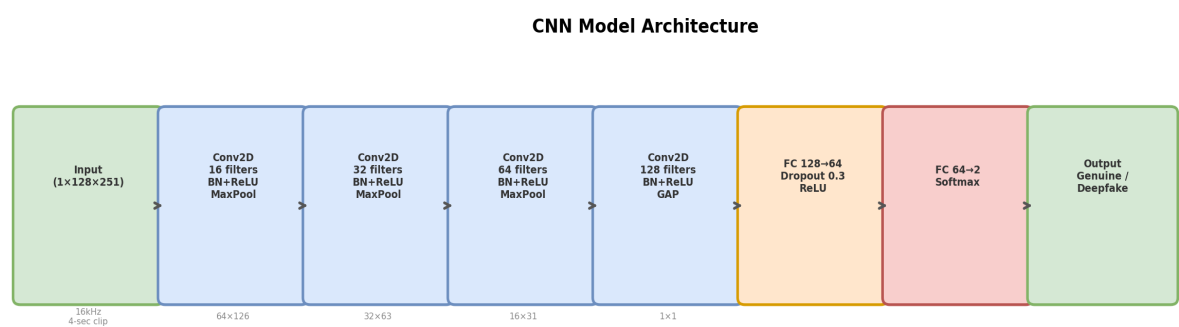


Figure: End-to-end CNN pipeline from raw audio to binary classification output.

6. Full Classification Report

Class	Precision	Recall	F1-Score	Support
Genuine	1.00	1.00	1.00	30
Deepfake	1.00	1.00	1.00	30
Accuracy			1.00	60
Macro avg	1.00	1.00	1.00	60
Weighted avg	1.00	1.00	1.00	60